

Exercise Sheet 3
Lecture “Cybersecurity”
Summer Semester 2026
*A Brief Summary
and
Modes of Operation of Block Ciphers*

Fundamentals

Übung 1.

- Kerckhoffs' principle plays a central role in modern cryptography. Restate it in your own words.
- In the past, cryptosystems were frequently designed whose security, contrary to Kerckhoffs' principle, relied on keeping the design secret. What is this approach called?
- The science of cryptology can be divided into the subfields of cryptography and cryptanalysis. Briefly describe both in your own words.
- Outline the classic scenario of two communication partners Alice and Bob communicating over an insecure channel. Use this to explain why communication partners must employ cryptographic algorithms.
- Match the variables and functions $e(\cdot)$, $d(\cdot)$, x , y , k , K and $|K|$, which are very important in cryptography, to their respective names. The options are: number of possible keys, key, encryption, key space, ciphertext, decryption and plaintext.

Adaptation of a Stream Cipher

Übung 2.

In a classical stream cipher, the plaintext x , the ciphertext y and the key stream s consist of individual bits x_i , y_i , s_i . Encryption and decryption are then defined as follows:

$$\begin{aligned}y_i &= e_{s_i}(x_i) = x_i \oplus s_i = x_i + s_i \pmod{2} \\x_i &= d_{s_i}(y_i) = y_i \oplus s_i = y_i - s_i \pmod{2}\end{aligned}$$

A stream cipher defined in this way can be very easily generalised so that it can operate on arbitrary alphabets and is no longer restricted to the binary alphabet. For hand ciphers, i.e. ciphers for manual encryption without a computer, it is for example advantageous if the stream cipher can be applied directly to letters.

- a) What alphabet can x_i , y_i , s_i take on in the function mentioned above?
- b) Suppose we now replace the binary alphabet with the Latin alphabet A, . . . , Z, where the letters are represented by the numbers 0, 1, . . . , 25. **How would the modulus m need to change relative to the function mentioned above?**
- c) What does the key stream look like now?
- d) What would need to change in the decryption function to keep it correct?
- e) Decrypt the following ciphertext HWHWZB with the key BSASRP.

A	0	F	5	K	10	P	15	U	20	
B	1	G	6	L	11	Q	16	V	21	
C	2	H	7	M	12	R	17	W	22	Z 25
D	3	I	8	N	13	S	18	X	23	
E	4	J	9	O	14	T	19	Y	24	

Tabelle 1: Caesar Alphabet, assigned to numbers (0-25)

Brute-Force Attacks

Übung 3.

AES is currently the most widely used symmetric cipher. In this exercise, the long-term security of AES with 128-bit keys¹ is examined. The assumption is that the best known attack is the exhaustive key search (also: brute-force attack), in which all possible keys are systematically tested. **Note:** *If you do not have a calculator, estimate your results using the laws of exponents and the assumption that 10^3 is approximately 2^{10} .*

- a) How many different 128-bit keys are there? Give the result as a power of two.
- b) We assume that the attacker has specialised hardware, so-called Application Specific Integrated Circuits (ASICs), which are optimised for AES key tests. One such ASIC can check $7 \cdot 10^8$ keys per second and the attacker has a budget of one million euros. A single ASIC costs 40€ and an overhead of 100% is assumed for the integration of the ASICs (building the computer, power supply, cooling, etc.). How many ASICs can be operated in parallel with the given budget? How long does an exhaustive key search take **on average**²? Put this time in relation to the age of the universe, which is approximately 10^{10} years.

¹A 128-bit number consists of 128 consecutive zeros and ones

²On average, only half of all possible keys need to be tested

Modes of Operation of Block Ciphers

In addition to the common block-by-block encryption (so-called **ECB** mode), there are further modes of operation for block ciphers, including the **CBC** and **OFB** modes. The ECB mode has several drawbacks, among others, uniform and large-scale regions that span multiple blocks in the ciphertext remain recognisable. As a result, the otherwise secure symmetric encryption in **ECB** mode can become insecure in a concrete application. An unfavourable scenario of encryption in **ECB** mode is illustrated in Figure 1. Figures 2, 3 and 4 show the individual modes of operation. The $\oplus$ operator represents a standard XOR operation.

Übung 4.

A simple block cipher $e(\cdot)$ with a 5-bit block size encrypts by performing a key-dependent bit permutation. For *one* specific key applied in this exercise, the block cipher performs the following simple encryption:

$$e(b_1b_2b_3b_4b_5) = (b_2b_5b_4b_1b_3)$$

Encrypt the message:

$$x = 01101\ 11011\ 11010\ 00110$$

using the respectively required mode of operation and state the ciphertext y in each sub-task.

- ECB**
- CBC** with IV = 11001
- OFB** with IV = 11001

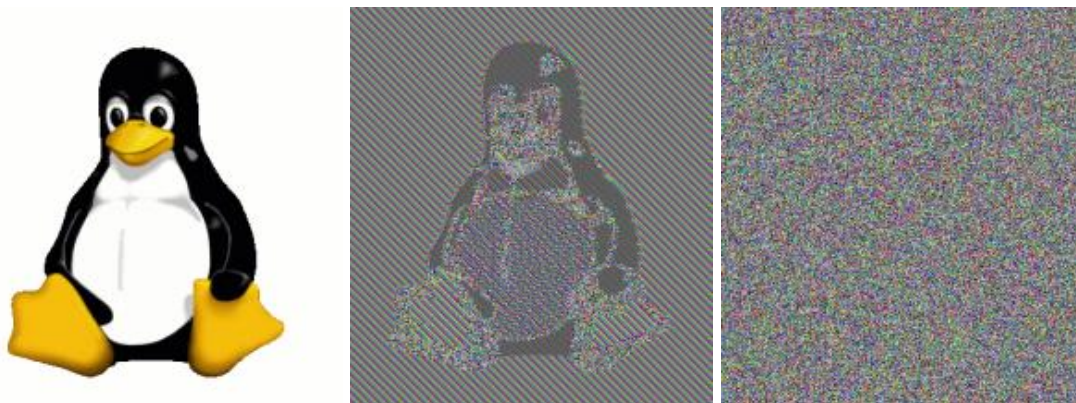


Abbildung 1: Example of a concrete unfavourable situation of **ECB** mode encryption

Left: Original image, **Centre:** Image encrypted in ECB mode, **Right:** Image encrypted in a chained mode

Image source: https://de.wikipedia.org/wiki/Electronic_Code_Book_Mode

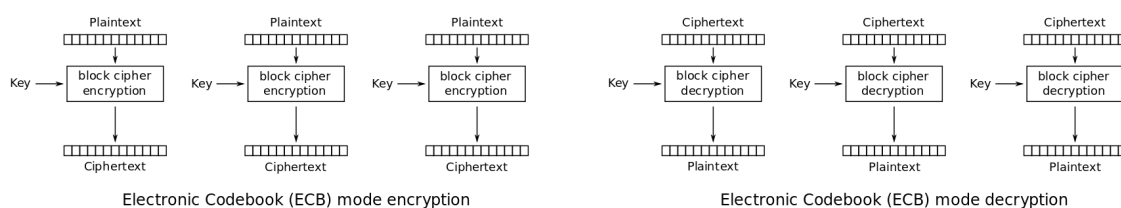


Abbildung 2: Encryption and decryption in **ECB** mode

Image source: https://de.wikipedia.org/wiki/Electronic_Code_Book_Mode

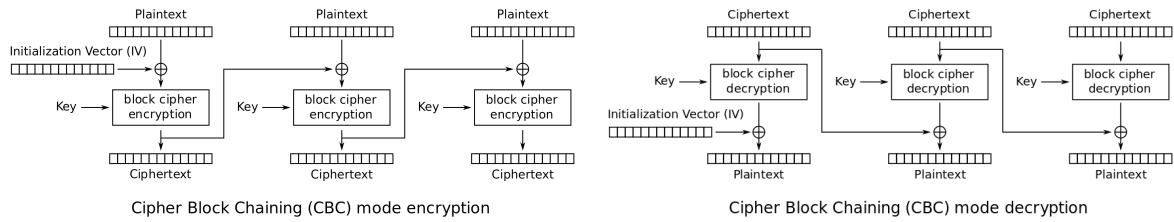


Abbildung 3: Encryption and decryption in **CBC** mode

Image source: https://de.wikipedia.org/wiki/Cipher_Block_Chaining_Mode

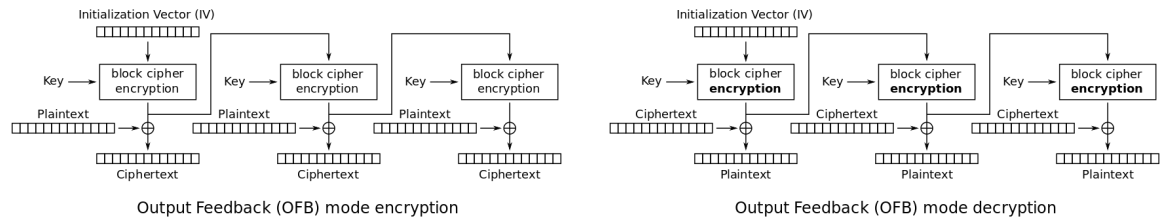


Abbildung 4: Encryption and decryption in **OFB** mode

Image source: https://de.wikipedia.org/wiki/Output_Feedback_Mode